MID TERM EXAMINATION

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 84 | Date: 14-03-2026

Total Questions: 18 | Answer all questions.

Name: ____________________________ **Roll No:** _________________________

Student Sign: ___________________ **Teacher Sign:** ___________________

SECTION A: MCQS

1. Which of the following best describes an Abstract Data Type (ADT)? **[2 Marks]**

a) A data structure that only stores abstract values.
b) A set of data values and associated operations, without specifying implementation details.
c) A concrete implementation of a data structure.
d) A data type that can only be accessed through pointers.

2. What is the worst-case time complexity for inserting an element at the beginning of a singly linked list? **[2 Marks]**

a) O(1)
b) O(log n)
c) O(n)
d) O(n log n)

3. A data structure where elements are added at one end and removed from the other end is known as a: **[2 Marks]**

a) Stack
b) Queue
c) Tree
d) Graph

4. Which of the following is a disadvantage of using arrays compared to linked lists? **[2 Marks]**

a) Faster access to elements by index.
b) Requires contiguous memory allocation.
c) More complex to implement.
d) Less memory efficient for a fixed size.

5. If a stack has a capacity of 5 elements and it currently contains 3 elements, what would be the result of a 'push' operation followed by a 'pop' operation? **[2 Marks]**

a) The stack size remains 3, and the pushed element is returned.
b) The stack size becomes 4, and the last pushed element is returned.
c) The stack size remains 3, and the element pushed before the last one is returned.
d) The stack size becomes 2, and an error occurs.

6. Briefly explain the purpose of Big O notation in algorithm analysis. **[3 Marks]**

SECTION A: SHORT ANSWERS

7. Define a Data Structure. **[3 Marks]**

8. State one primary advantage of a linked list over an array for dynamic data storage. **[3 Marks]**

9. Name two common applications of queues. **[3 Marks]**

SECTION A: LONG ANSWERS

10. What does the acronym "LIFO" stand for in the context of data structures? **[3 Marks]**

SECTION A: MCQS

11. Describe the steps involved in inserting a new node at the beginning of a singly linked list. **[5 Marks]**

SECTION A: SHORT ANSWERS

12. Differentiate between linear and non-linear data structures, providing an example for each. **[5 Marks]**

13. Explain how a Circular Queue addresses the limitation of "unutilized space" often encountered in a Linear Queue implementation using an array. **[5 Marks]**

14. List and briefly explain the primary operations on a Stack Abstract Data Type (ADT). **[5 Marks]**

SECTION A: LONG ANSWERS

15. Explain the concepts of Time Complexity and Space Complexity in the context of algorithm efficiency. **[5 Marks]**

SECTION A: MCQS

16. Design and explain the steps for implementing a basic Stack ADT using an array. Include the algorithms for push, pop, and peek operations. Clearly state any assumptions. **[10 Marks]**

17. Compare and contrast Arrays and Singly Linked Lists based on memory allocation, access method, insertion/deletion efficiency, and storage overhead. Provide scenarios where one is preferred over the other. **[12 Marks]**

SECTION A: SHORT ANSWERS

18. Explain the concept of a Doubly Linked List. Describe the algorithm for deleting a specific node (given its data value) from a Doubly Linked List. Illustrate with a conceptual diagram. **[12 Marks]**

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